Teacher's notes

OBJECTIVES

- Understand the concept of the volume of a solid shape.
- Understand the concept of the capacity of a container.
- Convert between units for capacity – millilitres, litres and kilolitres.
- Calculate the volume of a rectangular prism using the relationship between length, breadth, height and volume.
- Calculate the volume of various prisms using the formula $V = A \times h$.
- Calculate the capacity of a container.

PRIOR KNOWLEDGE

Students should be able to recognise and name common solid shapes. They should be able to perform the basic operations with whole numbers, fractions and decimals using mental arithmetic or a calculator.

BACKGROUND

Volume refers to the amount of space occupied by an object or substance. Capacity refers to the amount a container can hold. Capacity is only used in relation to containers.

The abbreviation cm^3 is read *cubic centimetre* and not 'centimetre cubed'.

STARTER ACTIVITY

Ask students to estimate and measure the volumes of various containers by filling them with cubes.

Build a cubic metre in the classroom. Brainstorm appropriate building materials and work with the class to construct it. You could discuss the fact that 1 000 000 cubic centimetres fit inside a cubic metre.

ACTIVITY SHEETS

4.1 Volume of solid shapes

This activity sheet develops students' recognition of the cubic centimetre as a unit for measuring volume. They are asked to determine the volume of a selection of solid shapes by counting cubes. Each cube has a volume of 1 cm^3.

Answers

1 a 4 cm^3 b 3 cm^3 c 4 cm^3 d 6 cm^3
e 8 cm^3 f 6 cm^3 g 12 cm^3 h 8 cm^3
i 18 cm^3

4.2 Volume of a rectangular prism A

This activity sheet develops the rule:
Volume = length × width × height.
Students start by counting the number of cubes in each layer. Then they multiply by the number of layers.

Answers

1 a 12 cm^3 b 36 cm^3 c 30 cm^3 d 36 cm^3
e 32 cm^3 f 125 cm^3

4.3 Volume of a rectangular prism B

In this activity sheet students are asked to calculate the volume of a number of rectangular prisms using the formula and the given dimensions. These diagrams are not drawn to scale. The activity sheet concludes with a selection of word problems. Students will need to answer these questions in their workbooks.

Answers

1 a 192 cm^3 b 1 344 cm^3 c 2 880 cm^3
d 1 694 cm^3 e 2 160 cm^3 f 5 760 cm

2 400 m^3 3 13.72 m^3 4 799 200 cm^3

4.4 Volume of a prism

This activity sheet develops the rule:
Volume = Area of face × height.
Students are then asked to calculate the area of a number of prisms using the given measurements. These diagrams are not drawn to scale.

Answers

1 a 300 cm^3 b 420 cm^3 c 576 cm^3
d 414 cm^3 e 672 cm^3 f 630 cm^3
g 1 300 cm^3 h 532 cm^3 i 1 728 cm^3
j 992 cm^3 k 540 cm^3 l 858 cm^3

2 a 144 cm^2, 2 592 cm^3 b 176 cm^2, 2 640 cm^3
c 384 cm^2, 5 376 cm^3 d 126 cm^2, 3 024 cm^3
e 72 cm^2, 1 656 cm^3 f 175 cm^2, 3 325 cm^3